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June 13, 2026

Editor-in-Chief
ACM Transactions on Mathematical Software

Dear Editor,

I am pleased to submit the enclosed manuscript, “*SCALPEL: A multi-backend FFT-NILT primitive for spectral slab PDE solvers, with class-dependent finite-precision auto-tuning*”, for consideration by ACM TOMS.

What is submitted. The manuscript presents SCALPEL, an open-source Apache-2.0 Python library that ships a single batched FFT-accelerated numerical inverse Laplace transform (NILT) primitive as a reusable kernel for spectral slab PDE solvers. The library targets a concrete gap in scientific Python: there is no portable, multi-backend, GPU-aware FFT-NILT primitive with built-in finite-precision parameter auto-tuning that downstream solvers can drop in without re-deriving the Bromwich-contour tuning. SCALPEL fills that gap, running unchanged across NumPy, PyTorch, JAX, and Julia on both CPU and GPU, with four physics-domain plugins shipped out of the box (lossy Maxwell, preparative chromatography, thermoviscous acoustics, and elastodynamics) plus a fractional Caputo single-term stress test.

What is new.

- A portable scientific-Python library exposing a batched FFT-NILT primitive at $\mathcal{O}(N_x N_y N_{\text{NILT}} \log N_{\text{NILT}})$ cost, with a twenty-five-callable public API spread across six modules: bundled physics propagators, memory-bounded chunked dispatch over the transverse grid, an introspection / diagnostics layer that exposes the auto-tuner’s decisions, a parameter-sweep harness, a validation API for matched-precision comparison against external references, and a configuration serializer for reproducible runs. A single-file dispersion-class plugin pattern adds new physics in ~ 80 – 150 lines per domain.
- A multi-backend dispatcher built on top of the Python Array API specification, supporting NumPy, PyTorch, JAX, and Julia (via PyJulia), with a documented subprocess sidecar pattern for benchmarking JAX CPU vs. GPU in one process.
- Class-dependent finite-precision parameter auto-tuning: two closed-form bounds on the largest transverse wavenumber the method can resolve at finite precision, one per dispersion class (telegrapher-type and diffusion-type), shipped as the `feasibility` module. Theorems 4.1 and 4.2 of the manuscript supply the derivation; users invoke a single API call.
- An eight backend $\times$ device wall-clock benchmark on a single workstation with a consumer GPU. Speedup over 3D Yee FDTD on the telegrapher class spans 8 – $153\times$ (high end on PyTorch CPU, GPU rows 27 – $121\times$); speedup over an explicit-time FTCS+L1 baseline on

fractional Caputo spans 12–394× (high end on CuPy GPU, remaining GPU rows 19–300×). Per-row data ship as a CSV in the reproducibility archive.

- A transform-domain head-to-head establishing that the speedup reflects FFT amortization across modes (the batched FFT-NILT is $\sim 5\times$ faster than an optimized de Hoog inversion and $\sim 13\times$ faster than fixed Talbot at matched accuracy on the dense-grid many-mode workload), not a comparison against weak baselines.
- Accuracy anchored against a `mpmath` 50-digit Talbot ground-truth, matching to 8×10^{-4} relative error at float64.
- A complete reproducibility pipeline (Apache-2.0 source, Zenodo archive, `CITATION.cff`, regression tests across all backends, one-command reproduction script, GitHub Actions CI matrix spanning three OSes $\times$ four Python versions plus per-backend extras, and a benchmark-methodology section in the article documenting matched-accuracy comparison and baseline parameter choices) compliant with the ACM Artifact Available checklist.

Why ACM TOMS specifically. The library is the contribution. The mathematical results (the dispersion-class structure, the finite-precision feasibility theorems) exist to back the parameter-selection module that ships with SCALPEL; they are not the lead claim. ACM TOMS is the natural venue for credit-allocation of numerical software with this depth of validation and reproducibility infrastructure — in particular, the manuscript is designed to map cleanly onto ACM’s Artifact Available badge requirements, with a filled-in checklist included in the reproducibility archive.

Reproducibility. All source, raw benchmark data (CSV), `mpmath` ground-truth reference data, figure-generation scripts, and the regression test suite spanning all four backends and the four physics domains are deposited at doi:10.5281/zenodo.20682437 under the Apache-2.0 license; a snapshot accompanies the submission. The ACM Artifact Available checklist is filled in at `reproducibility/acm_a`. We respectfully request consideration for the **ACM Artifact Available** badge.

Suggested reviewers. The following experts span the numerical-software and transform-domain communities the paper touches, and the author has no association with any of them within the past 48 months:

- **Bradley E. Treeby** (University College London, b.treeby@ucl.ac.uk) — lead author of *k*-Wave; expert on portable scientific PDE software with multi-platform support, and the natural reviewer for the acoustic-physics module.
- **Kristopher L. Kuhlman** (Sandia National Laboratories, klkuhlman@sandia.gov) — inverse-Laplace algorithms for Laplace-space numerical methods (Kuhlman 2013), and a natural reviewer for the NILT primitive and parameter auto-tuner.
- **Roberto Garrappa** (University of Bari, roberto.garrappa@uniba.it) — maintainer of the canonical Mittag-Leffler-evaluation library; the natural reviewer for the fractional Caputo stress test.
- **J. A. C. Weideman** (Stellenbosch University, weideman@sun.ac.za) — Talbot contour optimization for numerical inverse Laplace transforms (Weideman 2006), cited here as a fixed-Talbot baseline.
- **Christopher Rackauckas** (MIT / SciML, crackauc@mit.edu) — multi-backend scientific computing in Julia (DifferentialEquations.jl, SciML ecosystem); the natural reviewer for the multi-backend dispatcher architecture.

- **Stefan Behnel** (lxml; Cython) or another Array-API-consortium maintainer — portability layer expert who can speak to the Array API dispatch pattern.

The author has no preferred excluded reviewers.

Use of AI tools. The manuscript includes a statement on the use of AI tools, with author verification of all results, in accordance with ACM’s authorship policy; the author assumes responsibility for all content.

Competing interests. The author declares no competing interests.

Prior submission. This manuscript has not been published or submitted for publication elsewhere. The previously published Chem. Eng. Sci. paper (Vol. 328, 123776, 2026; doi:10.1016/j.ces.2026.123776) supplies the scalar NILT parameter-selection rules used by the auto-tuner and is cited as prior art.

Sincerely,

Gorgi Pavlov, PhD